

DAY 5

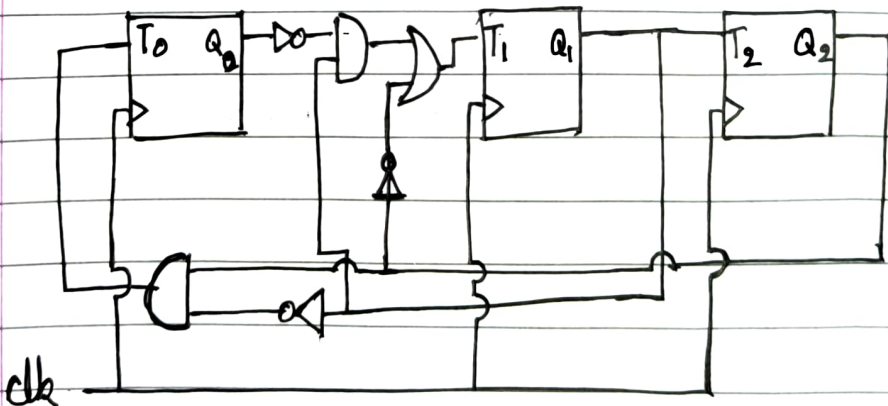
1.) Seq: $0 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 1 \rightarrow 0$
using 3 bit T-FF

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	0	1	0	1	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	1	1	1	0
1	1	0	x	x	x	x	x	x
1	1	1	x	x	x	x	x	x

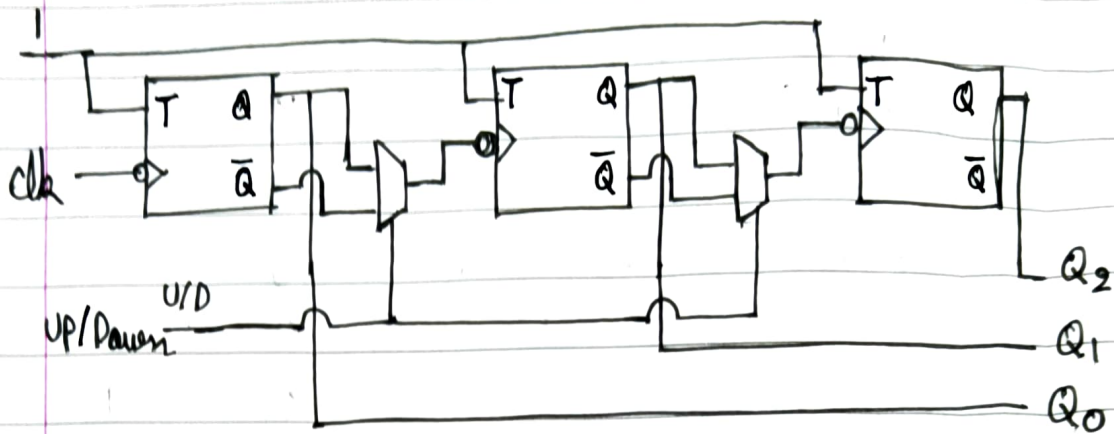
using K-MAP : $T_2 = Q_1$

$$T_1 = \bar{Q}_2 + Q_1 \bar{Q}_0$$

$$T_0 = Q_2 \bar{Q}_1$$



2.) negative edge Asynchronous 3bit



For up apply 0 at U/D ; ($clk = Q$)
For down apply 1 at U/D ($clk = \bar{Q}$)

3.) clk	Q_2	Q_1	Q_0	T_2	T_1	T_0	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	1	0	1	0
2	0	1	0	1	0	1	1	1	1
3	0	1	1	1	1	1	0	0	0

$\therefore 0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$